

# Coordinate Geometry Problem Set

Distance, Slope, and Game Development Applications

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**Instructions:** Assume a standard Cartesian coordinate plane. Show your work for each problem.

## Section A: Distance and Midpoint

1. Find the distance between points A(2, 3) and B(7, 15).
2. Find the midpoint of the segment joining C(-4, 6) and D(8, -2).
3. A point P(x, 4) is equidistant from A(1, 1) and B(5, 1). Find x.

## Section B: Slope and Lines

4. Find the slope of the line through (3, -2) and (-1, 6).
5. Write the equation of the line passing through (2, 5) with slope -3.
6. Find the equation of the line passing through (1, 2) and (4, 11), in slope-intercept form.

## Section C: Game Development Applications

Coordinate geometry underlies collision detection, movement, and camera logic in games.

7. **Sprite movement:** A character starts at position (50, 100) on screen and moves in a straight line to (200, 400) over 5 seconds at constant speed. What is the character's speed in pixels/second? (Use the distance formula.)

8. **Collision detection:** A circular enemy hitbox is centered at (300, 250) with radius 20 pixels. The player's position is (310, 265). Is the player within the enemy's hitbox range?

9. **Waypoint pathing:** An NPC patrols between waypoints (100, 100) and (100, 300), then (400, 300). If the NPC moves at a fixed speed, what is the total distance of one patrol loop (there and back through all three points)?

10. **Camera centering:** A game camera should stay centered on the midpoint between two players at (120, 80) and (480, 240), so both remain in view. Find the coordinates the camera should target.

11. **Line-of-sight:** A turret at (0, 0) can only fire in a straight line. An enemy is at (6, 8). What is the slope of the line the turret must aim along, and how far away is the enemy?